Monocular inspection of spacecraft under illumination constraints and avoidance regions

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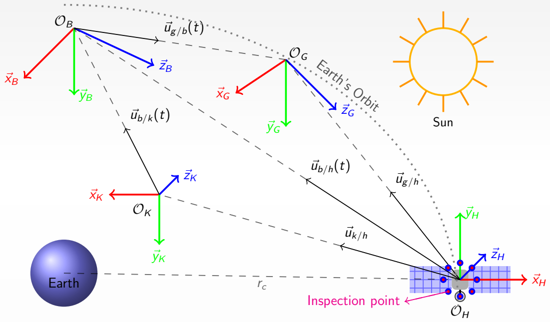
A deputy spacecraft is carrying out on-orbit inspection of a chief satellite while navigating relative to a chief satellite

Objectives

- Navigate the deputy spacecraft to a goal relative to the chief satellite.
- Use pixel coordinates of features H_i on the chief satellite and motion of the deputy spacecraft to localize the deputy spacecraft in the Hill frame.
- Maximize information gain from a camera fixed to the deputy spacecraft.
- Ensure the deputy spacecraft does not collide with or stray too far from the chief satellite.

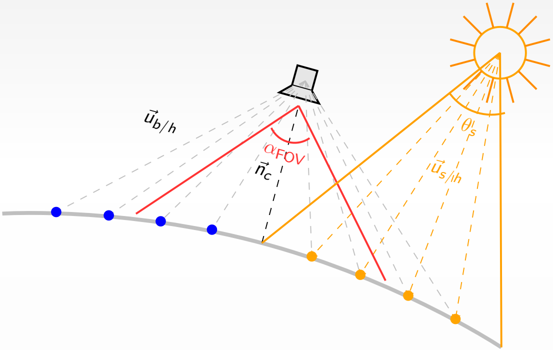


Schematic: Inspection problem in Hill Frame, $\mathcal{F}_B$ (Body frame), $\mathcal{F}_K$ (Keyframe), $\mathcal{F}_H$ (Hill frame), and $\mathcal{F}_G$ (Goal frame).

A deputy spacecraft is carrying out on-orbit inspection of a chief satellite while navigating relative to a chief satellite

Constraints

- Illumination Constraints - The Sun is the only light source in the environment.
- GPS Denied Navigation - Use onboard sensors for relative-localization.
- Field-of-View (FOV) Constraints - The deputy spacecraft has limited visibility and can only inspect features within its FOV that are illuminated by the Sun.
- Avoidance Regions - Avoid colliding with or drifting too far from the chief Satellite

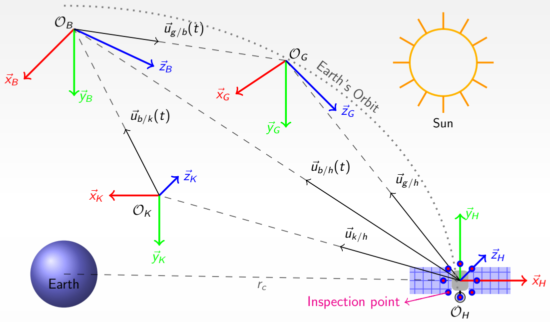


Visualization: The light emanating from the sun on the points on the surface of the chief satellite

Simplifying assumptions are needed to make the problem tractable

Assumptions

- The chief satellite is in a near-circular orbit around the Earth.
- The chief satellite and the deputy spacecraft start close to each other.
- The camera on the deputy spacecraft always faces the chief satellite.
- The rays of the sun are unobstructed from reaching the chief satellite.
- Orientation of the deputy spacecraft is fixed in the Hill frame.



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Motion of the deputy spacecraft relative to the goal can be modeled as a linear system

Linearized CW dynamics:

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t), \quad \mathbf{x}(0) = \mathbf{x}_0, \qquad (1)$$

State vector:

$$\mathbf{x}(t) := \begin{bmatrix} \vec{p}_{g/b}(t)^\top & \vec{v}_{g/b}^{\top}(t) \end{bmatrix}^\top \in \mathbb{R}^6,$$

Control vector:

$$\mathbf{u}(t) = \begin{bmatrix} (F(t))_x & (F(t))_y & (F(t))_z \end{bmatrix}^\top \in \mathcal{U}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix},$$

$$B = \frac{1}{m}\begin{bmatrix} 0_{3\times 3} \\ I_{3\times 3} \end{bmatrix}.$$

Relative navigation is formulated as an estimation problem

What is measurable?

- Directions to features H_i on the chief satellite $\vec{u}_{b/h_i}(t)$ and $\vec{u}_{k/h_i}(t)$ for $i \in \mathcal{N}$.
- The unit vector pointing from keyframe to the deputy spacecraft $\vec{u}_{b/k}(t)$.
- Position of G relative to H_i, $\vec{p}_{g/h_i}(t)$.
- Linear velocity of B relative to H $\vec{v}_{b/h}(t)$.

Estimation objective:

Estimate $\vec{p}_{g/b}(t)$, position of the goal relative to the deputy spacecraft.

Approach: Estimate r_{k/h_i}, use it to estimate $r_{b/h_i}(t)$ and $r_{b/k}(t)$. Then, use $\vec{p}_{g/h_i}(t)$, $\vec{p}_{h_i/k}(t)$, and $\vec{p}_{k/b}(t)$ to get $\vec{p}_{g/b}(t)$.

Dynamics of the unknown distances:

$$\dot{r}_{b/h_i}(t) = \vec{u}^{\top}_{b/h_i}(t)\vec{v}_{b/h}(t),$$

$$\dot{r}_{b/k}(t) = \vec{u}^{\top}_{b/k}(t)\vec{v}_{b/h}(t),$$

$$\dot{r}_{k/h_i} = 0.$$

Output equation:

$$\begin{bmatrix} \vec{u}_{b/h_i}(t) & -\vec{u}_{b/k}(t) \end{bmatrix} \begin{bmatrix} r_{b/h_i}(t) \\ r_{b/k}(t) \end{bmatrix} = \vec{u}_{k/h_i} r_{k/h_i}$$

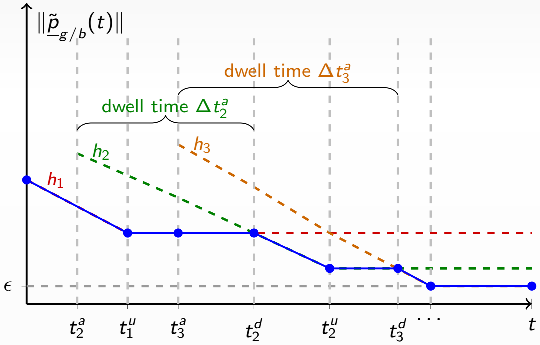
Features exiting the FOV of the deputy spacecraft necessitate a switched observer

Challenge: Features may not remain within the FOV of the camera long enough for convergence of $\hat{\vec{p}}_{g/b}$ to $\vec{p}_{g/b}$ within a desired error bound of $\epsilon > 0$.

Approach: daisy-chained distance estimates from multiple features.

Switched observer

- Compute $\hat{\vec{p}}_{k/h_1}(t)$ and then $\hat{\vec{p}}_{g/b}(t)$ using $\hat{\vec{p}}_{k/h_1}(t)$.
- When H_1 exits the FOV, freeze $\hat{\vec{p}}_{k/h_1}(t)$: $\hat{\vec{p}}_{k/h_1}(t) = \hat{\vec{p}}_{k/h_1}(t_1^u)$ for all $t \geq t_1^u$.
- Start estimating $\hat{\vec{p}}_{k/h_2}(t)$ whenever H_2 comes in the FOV.
- If $\left\|\tilde{p}_{k/h_2}(t)\right\| \leq \left\|\tilde{p}_{k/h_1}(t_1^u)\right\|$, start using H_2 to estimate $\hat{\vec{p}}_{g/b}(t)$.



Schematic: The switching logic for the switched observer.

Linear measurement equation enables the design of a memory-based observer

Observer ($\hat{\theta}_i(t) = \begin{bmatrix} \hat{r}_{b/h_i}(t) & \hat{r}_{b/k}(t) & \hat{r}_{k/h_i}(t) \end{bmatrix}^\top$):

$$\dot{\hat{\theta}}_i(t) = \begin{cases} 0, & \sigma_i = u, \\ \text{Proj}\left(\mu_i(t) + k_\theta \mathscr{Y}_i^\top(t)\left(\mathscr{U}_i(t) - \mathscr{Y}_i(t)\hat{\theta}_i(t)\right)\right), & \sigma_i = a, \end{cases}$$

$$\mu_i(t) = \begin{bmatrix} \vec{u}_{b/h_i}^\top(t)\vec{v}_{b/h}(t) \\ \vec{u}_{b/k}^\top(t)\vec{v}_{b/h}(t) \\ 0 \end{bmatrix}, \quad \mathscr{Y}_i(t) = \begin{bmatrix} Y_i(t) & 0_{3\times 1} \\ 0_{1\times 2} & 1 \end{bmatrix}, \quad \mathscr{U}_i(t) = \begin{bmatrix} \vec{u}_{k/h_i}\Sigma_{\mathcal{Y}_i}^{-1}(t)\mathcal{U}_i(t) \\ \Sigma_{\mathcal{Y}_i}^{-1}(t)\mathcal{U}_i(t) \end{bmatrix}, \quad \Sigma_{\mathcal{Y}_i} := \sum_{s=1}^{M} \frac{\mathcal{Y}_i^\top(t_s)\mathcal{Y}_i(t_s)}{1 + \|\mathcal{Y}_i(t_s)\|^2}, \quad \Sigma_{\mathcal{U}_i} := \sum_{s=1}^{M} \frac{\mathcal{Y}_i^\top(t_s)\mathcal{U}_i(t_s)}{1 + \|\mathcal{Y}_i(t_s)\|^2}$$

Observer error dynamics $\left(\tilde{\theta}_i := \theta_i - \hat{\theta}_i\right)$:

$$\dot{\tilde{\theta}}_i(t) = \begin{cases} 0, & \sigma_i = u, \\ -k_\theta \,\text{Proj}\left(\mathscr{Y}_i^\top(t)\mathscr{Y}_i(t)\tilde{\theta}_i(t)\right), & \sigma_i = a. \end{cases}$$

Uniform excitation condition: There exists a constant $\underline{\Sigma} > 0$, and a set of finite time instances $\mathcal{T} = \{t_i^*\}$ such that for all $t \geq t_i^*$ with $t_i^* \in (t_i^a,\ t_i^u)$, $\lambda_{\min}(\Sigma_{\mathcal{Y}_i}) > \underline{\Sigma}$. There exists a constant $\lambda_a > 0$ such that $\lambda_{\min}\left(Y_i^\top(t)Y_i(t)\right) > \lambda_a$ for all t.

Memory-based observer is exponentially stable under excitation conditions

Theorem (Analysis for a single feature)

Under assumptions, the Memory-based observer ensures that the trajectories of the concatenated observer error $\tilde{\theta}_i(\cdot)$ are globally exponentially stable such that for all i and for every pair of switching times (τ_i^u, τ_i^a), the trajectories of the observer error satisfies the bound

$$\left\|\tilde{\theta}_i(\tau_i^u)\right\| \leq \left\|\tilde{\theta}_i(\tau_i^a)\right\| e^{-\beta \Delta \tau_i^a}.$$

Corollary

Given a bound $\overline{\theta}_i^a$ such that $\|\tilde{\theta}_i(\tau_i^a)\| \leq \overline{\theta}_i^a$, the estimation error $\tilde{\theta}_i(t)$ at time $t = \tau_i^u$ is bounded as $\left\|\tilde{\theta}_i(\tau_i^u)\right\| \leq \overline{\theta}_{i-1}^u - \delta$, provided the feature stays in the FOV for a time $\Delta\tau_i^a = \tau_i^u - \tau_i^a \geq 0$ such that $\Delta\tau_i^a \geq -\frac{1}{\beta} \ln\left(\frac{\overline{\theta}_{i-1}^u - \delta}{\overline{\theta}_i^a}\right)$, where $\delta \geq 0$ is the desired margin of improvement from the previous distance estimate and $\overline{\theta}_{i-1}^u > 0$ is the bound on the observer error of the previous feature in the sequence.

Constraints to be enforced using a robustified barrier function

Collision constraint:

$$h_{\mathrm{KOZ}}(\mathbf{x}(t)) = \sqrt{2a_{\max}\left(\left\|\vec{p}_{g/b}(t) + \vec{p}_{g/h}\right\| - r_{\min}\right)} + \frac{\langle \vec{v}_{b/h}(t),\ \vec{p}_{g/b}(t) + \vec{p}_{g/h}\rangle}{\left\|\vec{p}_{g/b}(t) + \vec{p}_{g/h}\right\|} \geq 0.$$

Keep-in zone constraint:

$$h_{\mathrm{KIZ}}(\mathbf{x}(t)) = \sqrt{2a_{\max}\left(r_{\max} - \left\|\vec{p}_{g/b}(t) + \vec{p}_{g/h}\right\|\right)} + \frac{\langle \vec{v}_{b/h}(t),\ \vec{p}_{g/b}(t) + \vec{p}_{g/h}\rangle}{\left\|\vec{p}_{g/b}(t) + \vec{p}_{g/h}\right\|} \geq 0.$$

Robustified barrier function:

$$\Phi(\mathbf{x}(t), t) := \frac{\gamma_\Phi}{\frac{h_{\mathrm{KOZ}}(\mathbf{x}(t)) \cdot h_{\mathrm{KIZ}}(\mathbf{x}(t))}{h_{\mathrm{KOZ}}(\mathbf{x}(t)) + h_{\mathrm{KIZ}}(\mathbf{x}(t))} - L_h \epsilon_r e^{-\beta t}}, \quad \epsilon_r \geq \max_{r \in [r_{\min}, r_{\max}]} \left\{ \left\| r\vec{u}_{k/h_i}(0) - \vec{p}_{g/h}(0) - \hat{\vec{p}}_{g/b}(0) \right\| \right\}$$

Robustification ensures that in spite of estimation errors $\tilde{\vec{p}}_{g/b}(t)$, the deputy spacecraft does not collide with the chief satellite or stray too far from it.

The relative navigation problem is formulated as a constrained optimal control problem

Optimal control problem

$$\min_{\mathbf{u}\in\mathcal{U}} \int_0^{t_f} r\left(\mathbf{x}(\tau),\mathbf{u}(\tau)\right)\,\mathrm{d}\tau + V(\mathbf{x}(t_f),t_f),$$

$$\text{s.t.}\quad c(\mathbf{x}(t),\mathbf{u}(t),t)\leq 0,$$

where $r(\mathbf{x},\mathbf{u}) := \mathbf{x}^\top Q\mathbf{x} + \mathbf{u}^\top R\mathbf{u} + \gamma_c \left\langle C\mathbf{x}, \vec{n}_c\right\rangle^2$, is the running cost, V is the terminal cost, and $c(\mathbf{x},\mathbf{u},t) := \nabla_{\mathbf{x}}\Phi(\mathbf{x},t)\left(A\mathbf{x}+B\mathbf{u}\right) + \nabla_t\Phi(\mathbf{x},t) - \alpha(h_r(\mathbf{x},t))$, is the constraint function.

Orthogonality cost $\langle C\mathbf{x}(t), \vec{n}_c\rangle$

- $\vec{n}_c \in \mathbb{R}^3$ is the normal vector pointing outward from the centroid of the image-plane locations of the features currently within the FOV of the camera.
- Moving the camera orthogonally to $\vec{n}_c$ improves excitation.

T. E. Ogri, M. Qureshi, Z. I. Bell, K. Waters, and R. Kamalapurkar, “An adaptive optimal control approach to monocular depth observability maximization,” in Proc. Am. Control Conf., 2024

Closed-form solution to the optimal control problem

Optimal controller

$$\mathbf{u}^*(\mathbf{x}, t) = -\frac{1}{2}R^{-1}B^\top \nabla_{\mathbf{x}} \left(V^*(\mathbf{x}, t)\right)^\top - \frac{1}{2}R^{-1}B^\top \lambda^*(t)\nabla_{\mathbf{x}}\Phi^\top(\mathbf{x}, t).$$

A stabilizing controller obtained by solving a differential Riccati equation

$$\hat{\mathbf{u}}(\hat{\mathbf{x}}, t) = -R^{-1}B^\top P(t)\hat{\mathbf{x}} - \frac{1}{2}R^{-1}B^\top \hat{\lambda}(t)\nabla_{\hat{\mathbf{x}}}\Phi^\top(\hat{\mathbf{x}}, t),$$

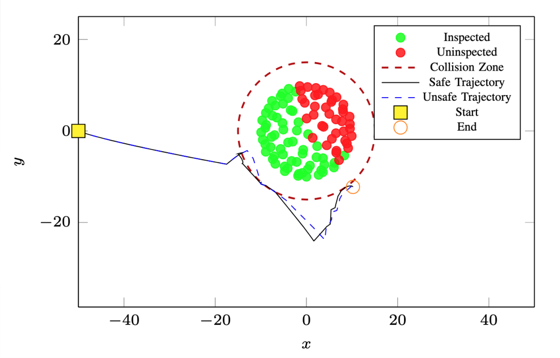
where $\hat{\lambda}(t) = \lambda^*(\hat{\mathbf{x}}, t)$ and $P(t)$ is the solution to the differential Riccati equation

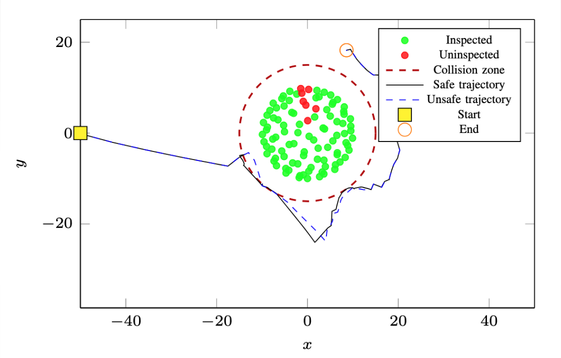
$$\dot{P}(t) = -A^\top P(t) - P(t)A + P(t)BR^{-1}B^\top P(t) - \underline{Q},$$

with the terminal condition $P(t_f) = Q_f + \gamma_c C^\top \vec{n}_c \vec{n}_c^\top C$.

Under sufficient gain conditions that depend on Lipschitz constants of the barrier function and the optimal Lagrange multiplier, we can show that trajectories that start close enough to the goal converge to a neighborhood of the goal.

Simulation results





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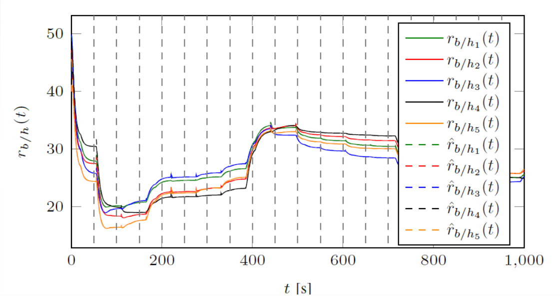

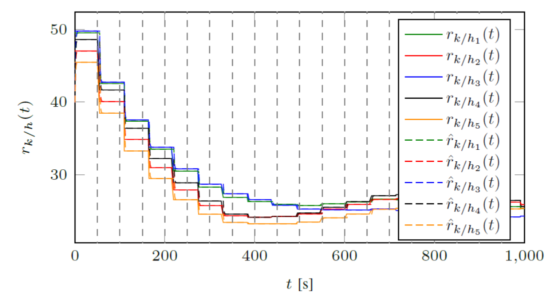

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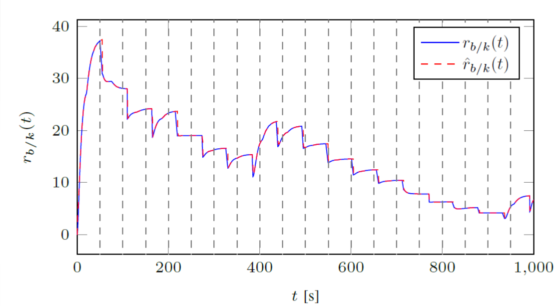
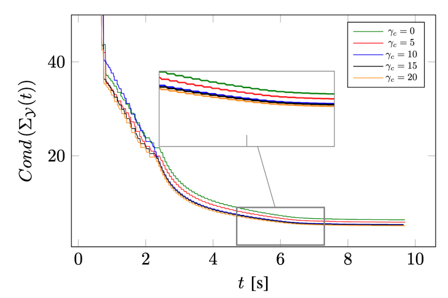
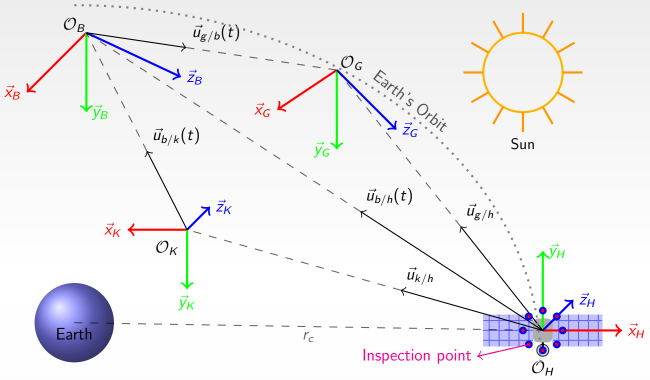




Limitations and future work

- The control design framework does not account for actuator saturation.
  - Given a set of constraints, find a subset that can be met under actuator saturation.
- Attitude dynamics are not considered in the current formulation.
  - Incorporate attitude dynamics into the optimal control formulation and design a safe attitude controller that enforces FOV constraints.


$\mathcal{O}_B$
$\vec{u}_{g/b}(t)$
$\vec{x}_B$
$\vec{y}_B$
$\vec{z}_B$
$\mathcal{O}_G$
Earth's Orbit
$\vec{x}_G$
$\vec{y}_G$
$\vec{z}_G$
Sun
$\vec{u}_{b/k}(t)$
$\vec{u}_{b/h}(t)$
$\vec{u}_{g/h}$
$\vec{z}_K$
$\vec{x}_K$
$\mathcal{O}_K$
$\vec{y}_K$
$\vec{y}_H$
$\vec{u}_{k/h}$
$\vec{z}_H$
$\vec{x}_H$
Earth
r_c
Inspection point
$\mathcal{O}_H$

Thank you!